Research Statement

Courtney Miller

I am fundamentally drawn to studying how developers navigate sociotechnical disruptions in software engineering. From digital infrastructure abandonment in open source software supply chains to generative AI (GenAI) tooling adoption on industrial development teams, ***I study the intersection of technical capabilities and human practices that determine successful transitions during periods of technological disruption***. These disruptions threaten the stability of critical digital infrastructure, including healthcare systems, financial services, and transportation networks. Addressing these disruptions is essential for ensuring software reliability, security, and the resilience of our broader digital economy.

My work is closely grounded in and guided by the needs of practitioners and the challenges they face as they emerge. ***I leverage rigorous multi-dimensional empirical mixed-method approaches, combining human-centered qualitative techniques with large-scale data-driven statistical analysis, modeling, and visualization, to understand the challenges developers face***, informing the design of theoretically-grounded insights and tooling solutions that address sociotechnical disruptions in software development and maintenance processes.

My research has been recognized by both the software engineering and open source communities. ***My first author conference publications have received four ACM Distinguished Paper Awards at the International Conference on Software Engineering (ICSE), and I am a recipient of the NSF GRFP.*** I have delivered invited talks at industry research venues (e.g., Linux Open Source Summit North America) and guest appearances on open source and academic podcasts (e.g., Sustain). I have contributed to 16 peer-reviewed publications at venues including ICSE and ESE/FSE, conducted large-scale studies analyzing millions of package dependency interactions [OSS'19], [ICSE'25], developed LLM-based classifiers that achieve high performance in independent practitioner evaluations [ICSE'26A], and led mixed-methods studies with a total of 3,259 developers (3,148 survey participants and 111 interviewees) [ICSE'21].

My dissertation focuses on the disruption of dependency abandonment; beyond my dissertation, my recent work investigates GenAI tooling adoption, which I argue is one of the most urgent disruptions developers face today. The remainder of this statement will focus on these two disruptions, followed by a discussion of my future research vision. ***As technology evolves, I will continue applying my approach and empirical expertise to support developers as we navigate the disruptions of tomorrow.***

1 Understanding Abandonment Disruptions

Modern software relies on open source as digital infrastructure, creating the software supply chains that our digital lives rely on. Nevertheless, many widely-used open source packages are abandoned by maintainers, exposing downstream users to the risk of unpatched vulnerabilities, targeted supply chain attacks, and costly unplanned migrations. This tension between expectations and reality has motivated the need to study and improve open source sustainability, though existing work has primarily focused on *preventing* abandonment.

My research takes a different approach: I study abandonment from the user perspective with the goal of enabling the sustainable use of open source digital infrastructure by helping users better navigate abandonment disruptions when they occur. By systematically studying abandonment as a disruption, we reveal developer challenges and inform the design of targeted solutions.

To understand developers' experiences with abandonment, we conducted an exploratory interview study with 33 developers discussing their experiences with abandonment [ESEC/FSE'23]. ***Developers struggle to respond due to limited resources and guidance. Many rely on manual labor-intensive investigations to identify abandonment that delay action.*** Often developers want to detect the abandonment of at least some dependencies before it triggered urgent problems. Thus, guided by signaling theory, we hypothesize that increasing information transparency surrounding abandonment can support responses.

Building on these insights, we conducted a large-scale quantitative statistical analysis measuring abandonment prevalence and downstream responses across the JavaScript npm ecosystem [ICSE'25]. We analyzed metadata for all packages in the npm ecosystem as of January 2015 (n=1,063,835) to identify the subset that were widely-used i.e., had at least 10,000 downloads in a month during our observation window (2015–2020).

We found that ∼15% of widely-used npm packages were abandoned during that window, yet only 18% of exposed projects removed the abandoned dependency. Additionally, users respond to abandonment at similar rates and with similar latency as they do to normal dependency updates (cf. Figure 1). To test our hypothesis about information transparency affecting remediation, we compared responses to explicit-notice abandonment (public end-of-support signals) with activity-based abandonment (sustained inactivity) using survival analysis. Projects facing explicit-notice abandonment were 58% more likely to remove abandoned dependencies, ***demonstrating that abandonment is under-addressed and improved transparency can significantly accelerate downstream response.*** We leverage these findings to inform the design of a prototype tool for identifying abandonment disruptions, which we discuss next.

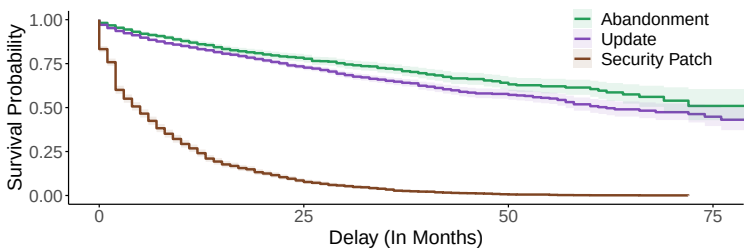


Figure 1: Survival probability for event "dependency event is not resolved" w.r.t. the date of event occurrence within dependent project's lifetime.

2 Improving Identification of Abandonment Disruptions

We learned in our interview study on abandonment that not all abandonment matters equally to many developers (cf. Section 1). As such, indiscriminate alerts could overwhelm developers and cause notification fatigue– a major usability issue limiting the effectiveness of existing tools for other dependency management practices e.g., vulnerability patching. While manual user customization is a potential solution, it is often not practically viable given the scale of modern dependency trees and the number of urgent tasks competing for developer's time. Thus, our key questions become (1) what abandonment *will* be impactful to a particular project given the context of their dependency usage; and (2) how to make such predictions at scale. We hypothesize and later demonstrate that our approach using ***large language models (LLMs), equipped with theory-driven reasoning and context-specific information, can accurately predict the impact of abandonment better than LLMs alone to support developer decision making*** [ICSE'26A]. This method is also promising for other disruptions where traditional tooling approaches have failed and theory or design work is still essential, as we will discuss further in the Future Work section.

We conducted need-finding interviews with 22 developers to develop a theoretical understanding of ***what factors influence the impactfulness of a dependency's abandonment on a project given the context of their dependency usage***, identifying four factors: *depth of integration*, *availability of alternatives*, *importance of the functionality provided*, and *external environmental pressures*. We then leveraged this theory to develop an LLM-based classifier to predict the project-specific impact of abandonment using theory-driven reasoning and context-specific information. We combine repository mining and static analysis using CodeQL to extract a body of project-specific usage information that would be sufficient for an expect to make a judgment of abandonment impact. Using the Retrieval-Augmented Generation (RAG) pattern, we embed this information in a theory-driven reasoning prompt, so the LLM can conduct similar reasoning and judgments. Finally, through an independent evaluation with 124 developers, we demonstrate that our classifier is effective at predicting project-specific impactfulness, achieving a Macro-F1 score of 0.840.

Beyond abandonment, I have investigated other critical disruptions in software supply chains. My work on maintainer disengagement reveals how burnout and life changes drive voluntary exits [OSS'19], informing retention strategies. My studies on toxicity [ICSE'22] and trust decomposition [ICSE-NIER'24] examine how social and security disruptions propagate through development communities. This broader perspective on disruptions enables more comprehensive and well-informed solutions.

3 Exploring Tomorrow's Disruptions: GenAI for Development

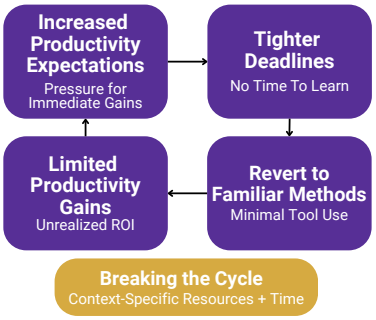
The rapid deployment of GenAI has disrupted software development on a seismic scale, promising a paradigm shift in how software is built and maintained. Yet despite widespread availability, adoption remains uneven even within teams, undercutting expected productivity gains, frustrating management expectations, and

casting uncertainty on the future roles of developers. As a first step, we ***explore why adoption varies among developers in seemingly similar contexts, utilizing a paired interview design with 54 developers at Microsoft representing 27 pairs from the same team who exhibit contrasting usage patterns (one frequent and one infrequent user)***, identified using telemetry data analysis [ICSE'26B]. This design ensures both developers share the same team-level context (e.g., codebase, manager, policies), ***enabling direct comparison of context-specific challenges and workarounds to yield an empirically grounded sociotechnical account of GenAI tool usage***. Our analysis reveals three key insights:

1. **Usage patterns diverge based on mindset and approach**: Frequent users often perceive GenAI as a collaborator, integrate it continuously and experimentally, and persist through challenges. Infrequent users often perceive it as a utility, apply it conservatively, and abandon it quickly when challenges arise.

2. **Organizational factors amplify adoption differences**: Team and organizational support interventions, e.g., team-specific demonstrations and knowledge sharing structures, can actively shape individual factors that impact technology adoption through an amplification effect.

3. **The Productivity Pressure Paradox**: Sufficient learning time appeared crucial for both initial adoption and advanced skill development, enabling the experimentation necessary to identify effective use cases in a particular workflow. Yet, many developers reported increased productivity expectations from management justified by tooling access without corresponding support, leading to pressure-induced reversion to familiar methods. This self-perpetuating dynamic hinders the skill development necessary to achieve expected productivity gains through effective tooling integration (cf. Figure 2).



These findings challenge the prevailing GenAI deployment strategy across the software industry, which delegates to individual developers responsibility for (1) determining how to use these tools to achieve expected productivity gains; and (2) navigating adoption-induced sociotechnical workflow disruptions, often while expecting maintained or increased velocity.

Figure 2: The Productivity Pressure Paradox.

Future Work

My long-term research vision centers on helping developers effectively navigate sociotechnical disruptions in software engineering. The software industry faces two major disruptions demanding immediate attention. First, GenAI adoption is fundamentally challenging established development processes. The impact of this urgently needs to be understood, particularly the long-term impacts on software quality, security, and maintainability. Second, software supply chains face escalating threats from dependency abandonment and sophisticated targeted attacks leveraging these weak links. I will address these disruptions through three complementary thrusts. First, I will develop AI-powered approaches that transform how we respond to supply chain disruptions, shifting from reactive crisis management to proactive risk management. Second, I will develop effective support interventions and infrastructure to facilitate GenAI adoption on development teams. Third, beyond adoption, I will help shape the evolution of software processes like code review for the GenAI era. ***My research recognizes that disruptions are not avoidable aberrations but inevitable evolutionary forces that require systematic study and strategic response.***

AI-Powered Supply Chain Sustainability and Security

I will develop three interconnected approaches that enable proactive risk management to address the unifying question: ***How can we leverage AI to transform supply chains into resilient ecosystems that anticipate and adapt to disruptions?***

Approach #1: Theory-Driven LLM Tools for Supply Chain Management. Given the success of using our theory-driven LLM-based approach to make context-specific predictions about the impact of abandonment disruptions (cf. Section 2), ***I will explore how we can adapt this approach to predict intelligent defaults for other supply chain security related tasks where the "right" answer depends on project-specific context and goals*** e.g., vulnerability triage prioritization. By systematically combining qualitative research to identify theoretical constructs with LLMs as scalable reasoning engines, this approach enables rapid interpretable tool development across diverse supply chain disruptions.

Approach #2: Project-Specific Response Blueprints. Developers emphasized that tooling support for identifying abandonment is not enough– many require actionable guidance on response strategies and next steps [ICSE'26A]. ***Can we develop project-specific blueprints that effectively guide developers through abandonment response?*** I will use participatory design to develop LLM-based tools for creating project-specific response blueprints, and user studies to measure their effectiveness at supporting responses.

Approach #3: Community-Oriented Response Strategies. Often, developers facing abandonment can reduce redundant collective response effort through community-oriented responses, e.g., posting migration guides, but often do not because of competing demands of their time, a form of the volunteer's dilemma [ESEC/FSE'23]. While literature suggests solutions for overcoming the volunteer's dilemma, their practical application in this context remains unexplored. Using empirical analysis of historical recovery patterns across hundreds of abandonment events, ***I will identify effective coordination strategies and develop evidence-based guidance for communities navigating infrastructure transitions.***

GenAI Integration Infrastructure

Our recent work on GenAI adoption (cf. Section 3) reveals obstacles to adoption at the organizational level, and my future work will study how to overcome them. ***How can we enable effective GenAI adoption and skill development using organizational support interventions and infrastructure?*** I will study how teams can adopt GenAI effectively by evaluating organizational interventions across diverse team contexts, designing integration infrastructure using participatory design and contextual inquiry, and developing metrics that capture meaningful skill development and value, not just usage. This work will help teams identify high-yield workflows and effective use cases, supporting sustainable GenAI adoption practices.

Software Process Evolution for the GenAI Era

GenAI-produced code disrupts established software development and maintenance process. ***My work will help shape process evolution for the GenAI era of software engineering***.

For example, modern code review assumes human-scale changes and comprehensible implementation logic. However, as the size of pull requests increases from hundreds to thousands of lines, developers are unable to perform the same comprehensive review process. ***How do code review practices evolve when reviews can't understand implementations?*** With collaborators, I have begun a large-scale quantitative analysis comparing code review practices across thousands of human versus GenAI authored pull requests, measuring defect detection rates, review comprehensiveness, and key security outcomes.

Agentic coding tools shift developers from direct implementation to intent-driven guidance, disrupting established development processes, and perpetuating the trend away from co-creation towards automation. ***How can agentic coding be leveraged effectively while preserving software security, compliance, and maintainability?*** In a recent preprint, we provide large-scale causal evidence of this tension: using difference-in-differences analysis of 800+ Cursor-adopting GitHub projects, we find that velocity gains are transient (dissipating within two months) while increases in code complexity and static-analysis warnings persist, creating a self-reinforcing cycle that dampens future productivity [MSR'26]. My research will investigate when and how agentic coding tools shift or erode developer capabilities.

Many developers in our GenAI adoption study (cf. Section 3) anticipated that GenAI could shift their role from lower-level implementation toward higher-level planning. In a complementary study of Project Managers (PMs) GenAI use that I collaborated on but have not discussed [ICSE-SEIP'26], we observed a parallel trend: Many PMs envisioned that GenAI could shift *their role* from higher-level planning to lower-level implementation. This symmetry signals a reconfiguration of roles on development teams in the GenAI era. ***My research will investigate how these roles evolve, how coordination and communication must adapt, and how we should rethink the makeup of a development team.***

References

[(OSS'19)] Courtney Miller, David Gray Widder, Christian Kästner, and Bogdan Vasilescu. "Why Do People Give Up FLOSSing? A Study of Contributor Disengagement in Open Source". In: *IFIP International Conference on Open Source Systems*. 2019, pp. 116–129.

[(ICSE'25)] Courtney Miller, Mahmoud Jahanshahi, Audris Mockus, Bogdan Vasilescu, and Christian Kästner. "Understanding the Response to Open-Source Dependency Abandonment in the npm Ecosystem". In: *Proc. Int'l Conf. Software Engineering (ICSE)*. 2025. ACM Distinguished Paper Award.

[(ICSE'26A)] Courtney Miller*, Hao He*, Weigen Chen, Elizabeth Lin, Chenyang Yang, Bogdan Vasilescu, and Christian Kästner. "Designing Abandabot: When Does Open Source Dependency Abandonment Matter?" In: *Proc. Int'l Conf. Software Engineering (ICSE)*. 2026. *Both authors contributed equally to this research.

[(ICSE'21)] Courtney Miller, Paige Rodeghero, Margaret-Anne Storey, Denae Ford, and Thomas Zimmermann. ""How Was Your Weekend?" Software Development Teams Working From Home During COVID-19". In: *Proc. Int'l Conf. Software Engineering (ICSE)*. 2021. ACM Distinguished Paper Award.

[(ESEC/FSE'23)] Courtney Miller, Bogdan Vasilescu, and Christian Kästner. ""We Feel Like We're Winging It:" A Study on Navigating Open-Source Dependency Abandonment". In: *Proc. Europ. Software Engineering Conf./Foundations of Software Engineering (ESEC/FSE)*. 2023.

[(ICSE'22)] Courtney Miller, Sophie Cohen, Daniel Klug, Bogdan Vasilescu, and Christian Kästner. ""Did You Miss My Comment or What?" Understanding Toxicity in Open Source Discussions". In: *Proc. Int'l Conf. Software Engineering (ICSE)*. 2022. ACM Distinguished Paper Award.

[(ICSE-NIER'24)] Lina Boughton, Courtney Miller, Yasemin Acar, Dominik Wermke, and Christian Kästner. "Decomposing and Measuring Trust in Open-Source Software Supply Chains". In: *Proc. Int'l Conf. Software Engineering: New Ideas and Emerging Results (ICSE-NIER)*. 2024.

[(ICSE'26B)] Courtney Miller, Rudrajit Choudhuri, Mara Ulloa, Sankeerti Haniyur, Robert DeLine, Margaret-Anne Storey, Emerson Murphy-Hill, Christian Bird, and Jenna L Butler. ""Maybe We Need Some More Examples:" Individual and Team Drivers of Developer GenAI Tool Use". In: *Proc. Int'l Conf. Software Engineering (ICSE)*. 2026. ACM Distinguished Paper Award.

[(MSR'26)] Hao He, Courtney Miller, Shyam Agarwal, Christian Kästner, and Bogdan Vasilescu. "Speed at the Cost of Quality: How Cursor AI Increases Short-Term Velocity and Long-Term Complexity in Open-Source Projects". In: *Proc. Int'l Conf. Mining Software Repositories (MSR)*. 2026.

[(ICSE-SEIP'26)] Mara Ulloa, Jenna L Butler, Sankeerti Haniyur, Courtney Miller, Barrett Amos, Advait Sarkar, and Margaret-Anne Storey. "Product Manager Practices for Delegating Work to Generative AI: "Accountability Must Not Be Delegated to Non-Human Actors"". In: *Proc. Int'l Conf. Software Engineering – Software Engineering in Practice Track (ICSE-SEIP)*. 2026.